Monetisation

Free to play with ads and four in-app products. The whole campaign can be finished without paying. Gameplay never calls a plugin: everything goes through the `Ads` and `Store` autoloads. Setup steps for the consoles: `BILLING_SETUP.md` (Play), `FIREBASE_SETUP.md`, and the AdMob section below.

Ads (Google AdMob)

Format	Where	Rules
Rewarded	Spin wheel: every spin after the daily free one ("WATCH AD: SPIN", up to `spin.ads_per_day` = 5)	Opt-in, button names the reward
Rewarded	Out of boulders: "WATCH AD: +1 Rocky"	Once per attempt. After that: Hellfire or Give up/replay
Rewarded	Win screen: "WATCH AD: +N coins" (doubles the level's coins)	Once per result screen
Rewarded	Out of energy: "WATCH AD" in the exhausted popup (placement `energy_refill`)	Pays for that one level; the next level needs rested bars
Interstitial	At the break after a match: the player taps **NEXT**, **RETRY** or **MAP** on the result screen, or **MAP** / **RESTART** in the pause menu	One per 4 matches (`ads.interstitial_every_matches`). A match is any attempt that ended, or was left or restarted after the first shot; wins and losses both count, training never. Count is saved across app restarts. Never during play, never before level 3, at least 60 s apart. If no ad is loaded the count is kept and it shows at the next break
Banner / app-open	Not used	They clash with the art and annoy players

- **Remove Ads** turns off every format. Rewards that used to need a video (extra spins, +1 Rocky, double coins) are then granted instantly, so buyers are never punished.
- Consent: Google UMP form before the first ad request (required in the EEA/UK).
- Plugin: Poing Studios godot-admob-plugin 5.1 (`addons/admob`), Gradle build.
- Debug builds use Google test units. Release builds read `res://ad_ids.cfg` (git-ignored): `[android] interstitial="ca-app-pub-XXXX/YYYY" rewarded="ca-app-pub-XXXX/ZZZZ"`
- The AdMob **App ID** is in project setting `admob/general/android/app_id`: `ca-app-pub-1292430265619610~9195063737` (real; debug builds still request Google's test units).
- Release ad units (in `ad_ids.cfg`): interstitial `ca-app-pub-1292430265619610/6882407348`, rewarded `ca-app-pub-1292430265619610/1846749261`.
- Max ad content rating: **T** (in code and in AdMob → Blocking controls). MA is blocked.

Why one interstitial per 4 matches

The owner asked for one ad per 4 matches, counting retries, not only new levels. The ad still appears only at a break the player chose (a results or pause button), so it never interrupts play, which is what the AdMob "unexpected interstitial" policy forbids. A

match counts once its first shot is fired, so opening and quitting a level does not count. Raise `interstitial_every_matches` if day-1 retention drops; lower it only after looking at Firebase data.

In-app products (Google Play Billing)

Product ID	Type	US price	Gives
`remove_ads`	Non-consumable (acknowledged, never consumed, restored automatically)	$4.99	No ads ever; ad rewards granted instantly
`coin_crate`	Consumable	$1.99	30,000 coins
`gem_crate`	Consumable	$4.99	400 gems
`hellfire_crate`	Consumable	$2.99	3 Hellfire strikes (purchase only)

Every crate has **fixed, shown contents** (no random loot boxes, so no odds disclosure or loot-box regulation applies). Prices shown in the game come from Google Play in the player's own currency; the US prices above are only fallbacks for the price templates. Regional prices: `BILLING_SETUP.md`.

Pricing rationale

- **Coin Crate $1.99 = 30,000 coins.** A free player earns about 120 coins per new star and 25 per replay, so 30,000 coins equals roughly 20-30 levels of play. It buys the Ballista (18,000) plus a few upgrades, or most of the way to the Trebuchet (40,000). Cheap enough to be an impulse buy.
- **Gem Crate $4.99 = 400 gems** = exactly one legendary skin (400). Free gems are deliberately scarce (boss first clears, a rare spin jackpot, daily day 7) and the total free gems in the game stay below one legendary skin, so gems stay special.
- **Hellfire Crate $2.99 = 3 strikes.** Hellfire guarantees a win on any level, including bosses. It is a skip, not power: free players can always replay a level. Three strikes at about $1 each is in line with paid "skip level" items.
- **Remove Ads $2.99**: the standard price point for this genre; low enough that ad-annoyed players convert.

Hellfire

Purchase only, never earned. When a level is lost because the boulders ran out, the popup offers "USE HELLFIRE" (if owned) or "BUY HELLFIRE x3". A strike drops a fire boulder on every block and defeats every archer (bosses too), so the level is always won. Tunables: `balance.hellfire`.

Purchase safety (see `ARCHITECTURE.md` → Billing)

Rewards only for `PURCHASED` state; pending purchases wait; cancelled/failed purchases show a message and grant nothing. Every purchase token is recorded in the save ledger before the consume call, so a token is never granted twice and a crash between "granted" and "consumed" is repaired on the next launch. An optional server verification hook is in `purchase_verifier.gd` (`balance.shop.verify_url`); no secrets in the client.

Currencies

Currency	Role	Earned	Spent on
Coins	Soft, plentiful	Wins (120 per first-clear star, 25 per replay win), boss first clears (1,500), daily rewards,	Siege engines (Ballista 18,000, Trebuchet 40,000),

Currency	Role	Earned	Spent on
		spin wheel, realm chests, achievements, double-coins video, Coin Crate	engine upgrade tracks, coin skins
Gems	Premium, scarce	5 per first boss clear, 3-gem spin jackpot, 5 on daily day 7, small achievement amounts, Gem Crate	Premium skins, coin exchange
Hellfire	Paid skip	Hellfire Crate only	Guaranteed win

Save version 4 removed blueprints (the parchment "map" icon). Old saves convert them to coins and refund old hero-upgrade spending.

Feature showcase

Players only buy what they know exists, so home shows a large "DISCOVER MORE" card now and then (`src/ui/promo_popup.gd`, rules in `UI_DESIGN.md` → Feature showcase, tunables in balance `promo`). It points to Remove Ads, Hero Addons, skins, crates, Workshop upgrades, siege engines and Special Missions. It never shows prices, timers or fake discounts, never opens during a level, skips anything already owned, and is capped at 4 a day. Measure it with `promo_shown` → `promo_tap` → `purchase_start`.

Policy rules we follow

- Ad buttons always say "WATCH AD" with a video icon and the exact reward; nothing looks like a game button that secretly opens an ad.
- No ads before consent, mid-level, on app open, on Retry, or in the first session.
- Prices are never hard-coded in the UI; they come from Play.
- No random paid rewards, no fake timers, no "only today" pressure.
- Full audit: `POLICY_AUDIT.md`.